

Intelligent Candidate Discovery

Team CULT

The Challenge

- Process 100,000 highly adversarial candidate profiles.
- Filter out 'keyword stuffers' and honeypot traps.
- Ensure deep semantic understanding of career histories.
- Execute ranking within strict < 5 minute, CPU-only limits.

Our Architecture

- A strict two-phase approach to guarantee performance constraints.
- Phase 1 (Offline): Heavy feature extraction, NLP embeddings, and hard filters.
- Phase 2 (Online): Lightning-fast arithmetic scoring and template-based reasoning.
- Cache Hand-off: Parquet file passing extracted signals between phases.

Phase 1: Offline Processing

- Text Embeddings: Used sentence-transformers/all-MiniLM-L6-v2.
- Focused heavily on 'career_history' descriptions rather than easily gamified 'skills' lists.
- Hard Filters: Automatically dropped job-hoppers and pure academic researchers.
- Honeypot Detection: Identified impossible expert skills with near-zero tenure.

Phase 2: Timed Ranking

- Hybrid Score: 30% Semantic Sim, 25% Demo Fit, 15% Exp, 10% Eval, -20% Penalties.
- Deterministic Reasoning: Generated factual sentence structures mapping to features.
- Zero Generative Hallucinations: No LLM APIs were called in the timed phase.
- Completed full 100k scale dataset well under the 5-minute requirement.

Conclusion

- Scalable: Easily processes millions of rows via batch embedding.
- Defensible: Every reasoning sentence maps cleanly to extracted profile data.
- Performant: Operates flawlessly on standard CPU hardware.
- Final Submission: Fully compliant with all Redrob Hackathon criteria.